

Math 421

Monday, October 13

Announcements

- Exam 1 Van Vleck B130
 - Wednesday 10/15 – 5:45-7:15pm
- This week's drop-in hours
 - **Tuesday** 9-11 am VV B205
 - Wednesday 12-1 pm **VV B205**
- No homework due Friday 10/17

Chapter 2: Sequences

Section 10 (Part 1): Monotone Sequences

Monotonic Sequences

Definition. A sequence (s_n) of real numbers is called **monotonic** if it is increasing or decreasing.

- A sequence (s_n) of real numbers is called an **increasing sequence** if $s_n \leq s_{n+1}$ for all n .
- A sequence (s_n) of real numbers is called a **decreasing sequence** if $s_n \geq s_{n+1}$ for all n .

Activity – Give an example of the following

- An increasing sequence - (n)
- A decreasing sequence - $(\frac{1}{n})$
- A non-monotonic sequence. - $(\sin n)$.

True/False

- ❖ Every convergent sequence is bounded.

T Theorem proved

- ❖ Every bounded sequence is convergent.

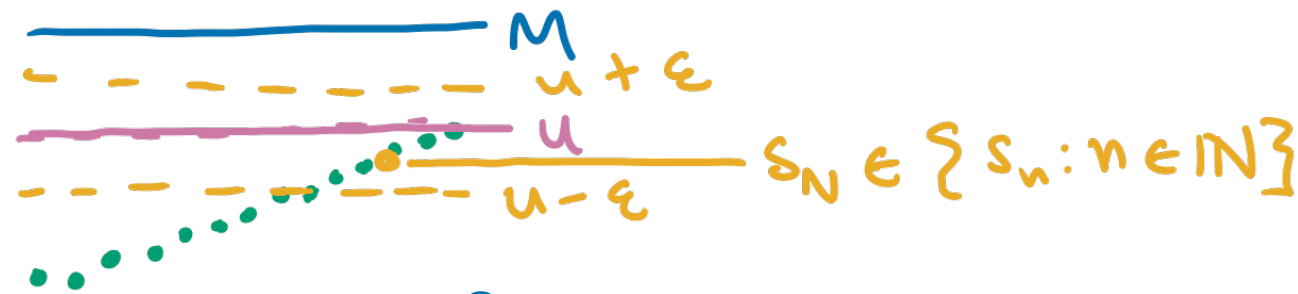
F $(-1)^n$

- ❖ Every convergent sequence is monotonic.

F $\left(\sin\left(\frac{n\pi}{2}\right) \left(\frac{1}{n}\right) \right)$ $\left(\frac{1}{n} (-1)^n\right)$

- ❖ Every monotonic sequence is convergent.

F (n)



Bounded Monotonic $\Rightarrow$ Convergent

Theorem 10.2. All bounded monotone sequences converge.

Discussion.

Case 1: Increasing
Case 2: Decreasing

- (s_n) bounded $\rightarrow \{s_n: n \in \mathbb{N}\}$ bounded $\rightarrow |s_n| \leq M, \{s_n: n \in \mathbb{N}\} \subseteq [-M, M]$
 $\hookrightarrow$ non empty & bounded above so $\sup \{s_n: n \in \mathbb{N}\} = u$

- $s_n \leq s_{n+1}$

Guess: $\lim s_n = u$

- $\lim s_n = s$. for all $\epsilon > 0$, there exists N such that
 $n > N, |s_n - s| < \epsilon$.

Bounded Monotonic $\Rightarrow$ Convergent

Theorem 10.2. All bounded monotone sequences converge.

Proof. Case 1: (s_n) increasing.

Let (s_n) be a bounded, increasing sequence. Since (s_n) is bounded, $\{s_n : n \in \mathbb{N}\}$ is non empty and since $s_1 \leq s_2 \leq s_3 \leq \dots$ the elements are increasing. By completeness, supremum of

$\{s_n : n \in \mathbb{N}\}$ exists and is a real number, u . Let $\varepsilon > 0$.

Since $u = \sup \{s_n : n \in \mathbb{N}\}$ there exists N such that $u - \varepsilon < s_N \leq u$. Assume $n > N$, since (s_n) is increasing

$$u - \varepsilon < s_N \leq s_n \leq u < u + \varepsilon$$

Thus, $|s_n - u| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary $\lim s_n = u$, so (s_n) converges.

* for monotonic sequences the limit always exists (i.e. converges or diverges to $\pm\infty$)

Unbounded Monotonic $\Rightarrow$ Diverges to Infinity

Theorem 10.4.

- i. If (s_n) is an unbounded increasing sequence, then $\lim s_n = +\infty$.
- ii. If (s_n) is an unbounded decreasing sequence, then $\lim s_n = -\infty$.

Proof of i. let (s_n) be an unbounded increasing sequence. $s_1 \leq s_2 \leq s_3 \leq \dots$
let $M > 0$. since (s_n) is unbounded $\{s_n : n \in \mathbb{N}\}$ is unbounded above. Hence there exists N such that $s_N > M$.
Assume $n > N$, then $s_n \geq s_N > M$. Since $M > 0$ was arbitrary, $\lim s_n = +\infty$.

Chapter 2: Sequences

Section 11 (Part 1): Subsequences

Subsequence

Definition. A **subsequence**, (t_n) , of a sequence, (s_n) , is a new sequence formed by selecting, in order, some (possibly all) of the terms in (s_n) .

Examples.

Let (s_n) be the sequence defined by $s_n = n^2(-1)^n$.

$$(-1, 4, -9, 16, -25, 36, \dots)$$

$$t_n = (4, 16, 36, 64, \dots) \quad \text{chose even terms from } (s_n) \quad t_n = s_{2n}$$

$$r_n = (-1, 4, -25, 64, \dots)$$

Activity

1. Let $a_n = 3 + 2(-1)^n$ for $n \in \mathbb{N}$.

- List the first four terms of the sequence (a_n) .

$(1, 5, 1, 5, \dots)$

- What terms could you select to produce a subsequence that is constant – i.e. the set of values is a single element?

even n or odd n .

2. Let $b_n = (-1)^n$ for $n \in \mathbb{N}$.

- What terms could you select to produce a subsequence that contains only positive terms?

- What terms could you select to produce the following subsequence $(1, 1, -1, -1, 1, 1, -1, -1, \dots)$?

$s_2, s_4, s_5, s_7, s_8, s_9, \dots$